

Corrected Eq. (29)–Eq. (41) Snippet for the Updated Shortcut Calculation

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Scope

This note records the corrected replacement formulae for the updated shortcut-to-absorbing-target calculation. The source checked is Luca's 2026-06-06 updated `Calculations.tex`. The leading shortcut resolvent sign has been corrected in the updated Eq. (29)–(31), but the correction has not been propagated consistently into Eq. (32), Eq. (33), Eq. (40), Eq. (41), and the final tail statement.

Let

$$\lambda = \beta(1 - q).$$

For a shortcut $u \rightarrow v$ where v is the absorbing target, the shortcut acts as killing in the transient state space: probability mass is removed from the $u \rightarrow u$ self-loop and absorbed at v .

Corrected generating-function chain

The updated Eq. (31) is the correct starting point:

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) - \frac{z\lambda\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 + z\lambda\tilde{W}_u(u, z)}. \quad (1)$$

Summing over transient sites and using

$$\sum_n \tilde{W}_r(n, z) = \frac{1 - \tilde{\mathcal{F}}_r(v, z)}{1 - z}$$

gives the corrected replacement for Eq. (32):

$$\boxed{\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z) \left[1 - \tilde{\mathcal{F}}_u(v, z)\right]}{1 + z\beta(1 - q)\tilde{W}_u(u, z)}}. \quad (2)$$

With

$$\tilde{H}_{N,|u-v|}(z) = \frac{\beta(1 - q)}{q} \frac{1}{1 + z\beta(1 - q)\tilde{W}_u(u, z)}, \quad (3)$$

the corrected replacement for Eq. (33) is

$$\boxed{\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + qz\tilde{W}_{n_0}(u, z) \left[1 - \tilde{\mathcal{F}}_u(v, z)\right] \tilde{H}_{N,|u-v|}(z)}. \quad (4)$$

For $N = 2L$ and $|u - v| = N/2$, the corrected H denominator is

$$\tilde{H}_{N,N/2}(z) = \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}. \quad (5)$$

This plus sign is consistent with (3).

Corrected time-domain convolution

The corrected replacement for Eq. (40) is

$$F_{n_0}(v, t) = \mathcal{F}_{n_0}(v, t) + q \sum_{t'=0}^{t-1} W_{n_0}(u, t-1-t') \sum_{t''=0}^{t'} [\delta_{t', t''} - \mathcal{F}_u(v, t' - t'')] H_{N, N/2}(t'') \quad (6)$$

for $t \geq 1$, with $F_{n_0}(v, 0) = 0$ when $n_0 \neq v$.

Clean replacement for Eq. (41)

The long mixed α, γ, s convolution form is not the safest final form. For the symmetric antipodal case, set $N = 2L$, $v = 0$, $u = L$, and let $\rho \in \{0, \dots, L\}$ be the folded distance from the initial site to the target. Define

$$a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1/z - 1}{q}, \quad U_{-1}(y) = 0.$$

The corrected first-passage generating function reduces to

$$\tilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}. \quad (7)$$

Let

$$D(y) = aT_L(y) + U_{L-1}(y), \quad N_\rho(y) = aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y).$$

If $D(y_k) = 0$ and $s_k = 1 - q + qy_k$, then for simple roots

$$F_\rho(t) = \sum_k B_{\rho k} s_k^{t-1}, \quad B_{\rho k} = \frac{q N_\rho(y_k)}{D'(y_k)} \quad (8)$$

for $t \geq 1$. This is the recommended time-domain replacement for Eq. (41).

Corrected tail statement

The apparent α_ℓ and γ_r poles in the long convolution expansion are intermediate poles. In the closed form (7), the ultimate tail is controlled by the largest positive root s_1 of $D(y) = 0$:

$$F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad \frac{F_\rho(t+1)}{F_\rho(t)} \rightarrow s_1. \quad (9)$$

Equivalently,

$$\tau_\beta^{-1} = -\log s_1, \quad aT_L \left(1 + \frac{s_1 - 1}{q} \right) + U_{L-1} \left(1 + \frac{s_1 - 1}{q} \right) = 0. \quad (10)$$

For weak shortcut strength,

$$s_1 = \alpha_1 - \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2), \quad 1 - s_1 \sim \frac{q\pi^2}{2N^2} + \frac{2\beta(1-q)}{N}. \quad (11)$$

Thus the generic $t\alpha_1^t$ tail in the updated manuscript should be removed.

Numerical checks

For $N = 100$, $q = 2/3$, $\beta = 0.04$, shortcut $u = 6 \rightarrow v = 56$, and target $v = 56$, direct finite-matrix checks give:

Formula compared with the direct stochastic matrix	max absolute error
correct plus/plus first-passage formula	2.08×10^{-17}
updated Eq. (32) denominator sign	3.29×10^{-3}
updated Eq. (33) minus prefactor	6.34×10^{-2}
closed form (7)	7.63×10^{-17}

For the same $N = 100$, $q = 2/3$, the proposed equality

$$s_1 = 1 - q + q \cos \frac{2\pi}{N}$$

does not occur for finite positive β in the admissible range. The corrected dominant pole satisfies

$$\gamma_1 < s_1 < \alpha_1, \quad \gamma_1 = 1 - q + q \cos \frac{2\pi}{N}, \quad \alpha_1 = 1 - q + q \cos \frac{\pi}{N}.$$

Under the clear visual double-peak criterion used in the local scan, clear double peaks occur up to about $\beta = 0.03$ for $n_0 = 1, 2, 3$, and no clear double peak remains at $\beta_c = 2q/[(1-q)N] = 0.04$.